

Overview of BigQuery's schema editing abilities

Previously, you learned about schemas, which are a specific type of data model used in relational databases. They define the structure of the tables, including the columns and the data types. You also explored some of the basics for table schemas in BigQuery. Throughout this program, you'll be using BigQuery as a way to practice your SQL writing skills, and querying a database. BigQuery has schema editing capabilities that will help you design and modify table schemas quickly. In this reading, you'll explore some of these functions so that you can start working with schemas in your own BigQuery projects.

Schema auto detect and advanced options

The BigQuery schema auto-detection function enables BigQuery to intuit the schema for your table. This includes identifying the data types for each column by selecting a sample, and analyzing the value to assign its data type. Auto-detection works for CSV, JSON, or Google Sheets data. You won't need auto-detect for Avro, Parquet, ORC, Firestore export, or Datastore. These files are self-describing, and contain their schema definition within their contents, which BigQuery can use. Along with detecting schema details, auto-detect can also identify file compression, and date time values. You can find the auto-detect option on the Create table menu.

The screenshot shows the 'Create table' dialog box in BigQuery. It is divided into two main sections: 'Destination' and 'Schema'.

Destination section:

- Project ***: learning-labs-sandbox (with a 'BROWSE' button to the right)
- Dataset ***: Examples
- Table ***: Automobile data (with a note below: 'Unicode letters, marks, numbers, connectors, dashes or spaces allowed.')
- Table type**: Native table (with a dropdown arrow and a help icon)

Schema section:

- ☒ Auto detect

At the bottom, there are two buttons: 'CREATE TABLE' (in blue) and 'CANCEL'.

In addition to the auto detect function, the Create Table menu also has advanced options so that you can also easily apply partitions and clustering to your table.

Partition and cluster settings

Partitioning

No partitioning

Clustering order

Clustering order determines the sort order of the data. Clustering can be used on both partitioned and non-partitioned tables.

Advanced options

Write preference

Write if empty

Number of errors allowed

0

☐ Unknown values

Field delimiter

Comma

Quote character

Double quote

Header rows to skip

0

Modify existing schemas

BigQuery allows you to easily modify existing schemas while previewing your changes, so you can intuitively update your table schemas. This means you can add or remove columns, or change a column's name, type, or default mode. To access the schema editor, select the table you want to modify the schema for. Under the **Schema** menu, you'll have the option to **Edit the schema**.

Automobile data

QUERY

SHARE

COPY

SNAPSHOT

DELETE

EXPORT

REFRESH

SCHEMA

DETAILS

PREVIEW

TABLE EXPLORER

LINEAGE

DATA PROFILE

DATA QUALITY

Filter

Enter property name or value

	Field name	Type	Mode	Key	Collation	Default Value	Policy Tags	Description
☐	make	STRING	NULLABLE					
☐	fuel_type	STRING	NULLABLE					
☐	num_of_doors	STRING	NULLABLE					
☐	body_style	STRING	NULLABLE					
☐	drive_wheels	STRING	NULLABLE					
☐	engine_location	STRING	NULLABLE					
☐	wheel_base	FLOAT	NULLABLE					
☐	length	FLOAT	NULLABLE					
☐	width	FLOAT	NULLABLE					
☐	height	FLOAT	NULLABLE					
☐	curb_weight	INTEGER	NULLABLE					
☐	engine_type	STRING	NULLABLE					

EDIT SCHEMA

VIEW ROW ACCESS POLICIES

From there, you can edit or remove existing columns:

Edit Schema

Current schema

ADD POLICY TAG

Filter Enter property name or value

☐	Field name	Type	Mode	Collation	Default Value	Policy Tags	Description
☐	make	STRING	NULLABLE		Default Value		Description
☐	fuel_type	STRING	NULLABLE		Default Value		Description
☐	num_of_doors	STRING	NULLABLE		Default Value		Description
☐	body_style	STRING	NULLABLE		Default Value		Description
☐	drive_wheels	STRING	NULLABLE		Default Value		Description
☐	engine_location	STRING	NULLABLE		Default Value		Description
☐	wheel_base	FLOAT	NULLABLE		Default Value		Description
☐	length	FLOAT	NULLABLE		Default Value		Description

SAVE **CLOSE**

Or add new ones:

New fields

☐ Edit as text

1 **Field name *** average_price **Type *** NUMERIC **Mode** NULLABLE **Precision** **Scale** **Rounding Mode** ROUNDING_MODE_UNSP... **Description**

2 +

SAVE **CLOSE**

Key takeaways

Schemas are a necessary part of working with relational databases; they describe the data, and how it's connected. Without a schema, querying data becomes impossible, and the database can't interpret the data or know how to store it. Luckily, BigQuery is already equipped with tools like schema auto-detection and schema modification that can make working with your table schemas that much easier. This will be useful for working with BigQuery later in this certificate program.